

Shashank Chakilam

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EDUCATION

Masters of Science in Electrical & Computer Engineering, San Francisco State University	Jan 2023- May 2025 GPA: 3.8
Bachelor of Engineering in Electronics & Communication Engineering, JNTUH	Jul 2017- Jul 2021 GPA: 3.8

SKILL

Programming Languages: SystemVerilog, Verilog, Python, Assembly Language, Java, C, C++.

Technologies: UVM, Digital Design, VLSI Design, RTL Design, Computer Architecture, OOPs, CMOS, Circuits, SoC, ASIC.

EDA Tools: Synopsys Design Compiler, PrimeTime, VCS, Cadence Innovus.

PCB Design Tools: KiCad, ECAD (Electronic Computer-Aided Design).

Protocols: AMBA, PCIe, USB, DDR, I2C, SPI.

Development Boards: Raspberry Pi, BeagleBone, Arduino (Uno, ATmega-328, NodeMCU ESP-8266), PSoC6.

Operating Systems: Windows, Linux, macOS, Unix.

EXPERIENCE

Automation Engineer, Netcracker Mar 2021 - Dec 2022

- Developed and executed **automated test scripts** using **Python and TCL**, improving **testing efficiency by 30%**.
- Designed and implemented **test frameworks**, reducing **manual effort by 40%** and accelerating the **release cycle**.
- Collaborated with cross-functional teams to identify and resolve defects, ensuring **high-quality product delivery**.
- Optimized **legacy automation scripts**, reducing **test execution time by 40%** and cutting **manual testing efforts**.
- Utilized **Selenium** and other tools to create scalable test automation solutions, reducing **test failures by 35%**.
- Gained hands-on experience with **Linux/Unix environments** and version control systems like **Git**.

Application Developer, Zetatek Industries May 2019 - Jul 2019

- Developed an **industrial machine control application** using **LabVIEW**, ensuring seamless functionality and performance.
- Gained hands-on experience in **industrial automation and control systems**, enhancing technical proficiency in real-world applications.

Research and Development Engineer, KMIT May 2018 - May 2021

- **Humanoid Robot with Speech and Gesture Recognition**
 - Developed a **Humanoid Robot** capable of speech, head movements, handshakes, and gestures to identify individuals and assist with college inquiries, enhancing campus interaction.
 - Integrated **Alexa** for student information access, **machine learning algorithms** for face recognition, and **servo motor controls** for coordinated movement.
 - Utilized **Raspberry Pi** for processing and **Arduino** for motor control, achieving seamless hardware-software integration.
- **GUI-Based Control System for Test Tube Spinning Machine**
 - Developed a **GUI-based control system** for a **Test Tube Spinning Machine**, streamlining user interactions and improving operational efficiency.
 - Designed an intuitive interface to control machine parameters, ensuring ease of use and precision.

PROJECTS

Verification of Fixed-Point Arithmetic Design using System Verilog

- Designed and implemented a System Verilog based testbench for a fixed-point arithmetic design, achieving **over 90% functional coverage** through constrained-random stimulus and functional coverage models.
- Improved **bug detection rate by 30%** and reduced **verification time by 35%** using Python and TCL scripting for automation.
- Developed SystemVerilog assertions (SVAs) and test cases to ensure design compliance and reliability.

ASIC Implementation of Motion Estimator in 14nm CMOS using Verilog

- Achieved **timing closure** and reduced **design size by 40%**, meeting all timing, area, and power constraints within project deadlines.
- Led **RTL design and verification**, developing testbenches and performing synthesis using **Synopsys Design Compiler** and timing analysis with **Synopsys PrimeTime**.

- Managed **physical design** to ensure compliance with 14nm CMOS requirements, optimizing for performance and power efficiency.

Static Timing Analysis of a Chip using Synopsys

- Performed **full-chip timing analysis** using **Synopsys PrimeTime**, ensuring **timing closure** across multiple corners (setup, hold, and transition).
- Identified and resolved **critical timing violations** by optimizing clock tree synthesis (CTS) and buffer insertion.
- Generated and analyzed timing reports, providing actionable recommendations to improve **slack margins** and meet design specifications.
- Collaborated with the design team to implement fixes, achieving **sign-off** with zero timing violations.